

Molecular transport in ionic liquids under confinement studied by low field NMR

Carlos Mattea^{a,*}, Amin Ordikhani-Seyedlar^{a,1}, Patrícia Kelen de Almeida Silva^b, Siegfried Stapf^a

^a Institute of Physics, Ilmenau University of Technology, D-98693 Ilmenau, Germany

^b Universidade Federal de São João del Rei, MG 36301-160 São João Del Rei, Brazil

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ABSTRACT

Films from aqueous gelatin solutions following the addition of several types of ionic liquids were studied. The water was allowed to evaporate until the films reached their final steady condition, while the ionic liquids remained in the matrix structure because of their negligible vapor pressure. Transversal and longitudinal relaxation times and diffusion coefficient were determined with the help of a single-sided mobile nuclear magnetic resonance scanner. Conductivity measurements were performed in addition. The results reveal different dynamics of the ionic liquids under confinement in the gelatin film porous matrix. The films become rigid and opaque using anions containing Fluor, while they are more flexible and semitransparent, when dicyanamide anion is used. The effect of the geometrical restrictions of the gel matrix on the dynamics of the solvent is more pronounced for the case of the flexible films, while the conductivity remains close to the bulk.

1. Introduction

The development of materials based on polymeric matrices combined with room temperature ionic liquids (RTILs) has gained considerable interest during recent years due to their applications as electrolytes in conducting films [1] with application in batteries and other related energy-storage systems. The functionality and properties of these materials can be adjusted using different types of RTILs and hosting polymers. In particular, their combination with biopolymers expands these applications towards the field of biosensors [2]. The knowledge of dynamical and physicochemical properties in these systems is necessary in order to characterize the microscopic transport properties of the mobile ions responsible for the electrical conductivity [3,4,5].

Gelatin is a water-soluble denatured protein obtained by the partial hydrolysis of collagen. During film preparation, the gelatin in solution with water is first heated at $\sim 50^\circ\text{C}$, and later cooled below the coil/helix transition temperature (40°C). In this process, some of the molecular chains conforming the gelatin combine with each other to form a partial collagen-like triple helix. The combination of dicyanamide-based ILs with gelatin have been used recently to prepare conducting films [6].

Single-sided mobile NMR scanners [7] allow the study of films, both during formation in an open environment [8,9], and after their

formation process is completed. The high field gradient strength of the devices permits the discrimination between the two different nuclei ^1H and ^{19}F , changing the relative position of the sample respect to the magnet [10] (see Fig. 1). This is particularly suitable because in many RTILs, ^1H and ^{19}F are found in different ions.

In this contribution, ILs and water are used as solutions to dissolve gelatin and prepare films. Due to the low volatility of ILs, they will remain in the final film, while the water evaporates with only a small irreducible water fraction remaining. Arising questions are: What is the mobility of the confined IL in comparison to their bulk values? Could the film be used as a conducting film? The focus is put on the characterization of the dynamics of the ILs and trying to answer those questions.

2. Materials and methods

The RTIL 1-butyl-3-methylimidazolium dicyanamide ($\text{Bmim}^+\text{DCA}^-$), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ($\text{Emim}^+\text{Tf}_2\text{N}^-$), 1-Ethyl-3-methylimidazolium tetrafluoroborate ($\text{Emim}^+\text{BF}_4^-$), were used. The detailed procedure of the preparation of the films are given in [Supplementary Contents](#).

For the sample Gel-EmimBF₄, 1 ml of EmimBF₄ was used. In the case of the Gel-EmimTf₂N, two samples were prepared, one containing 1.5 ml of EmimTf₂N (1:1.5) and the other one, 2.5 ml of EmimTf₂N

* Corresponding author.

E-mail address: carlos.mattea@tu-ilmenau.de (C. Mattea).

¹ Current address: Dept. of Physical Chemistry, University of Paderborn, D-33098 Paderborn, Germany.

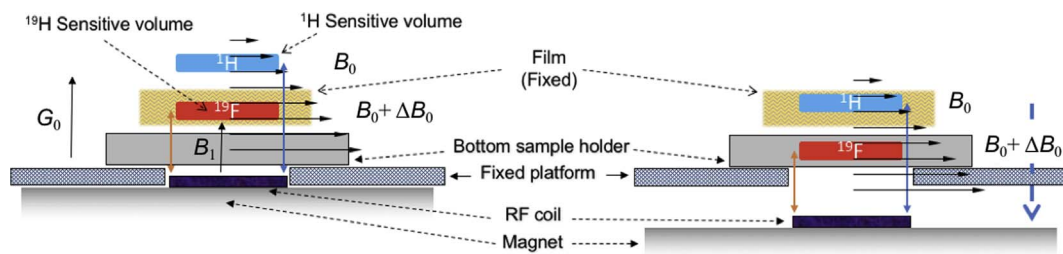


Fig. 1. Scheme of the single-sided NMR set-up showing both, the sensitive regions corresponding to the ^1H and ^{19}F resonance. The sensitive volumes remain at a fixed distance from the surface of the magnet, which can be moved in the vertical direction (B_0 , magnetic field; B_1 , radiofrequency field; G_0 , magnetic field gradient). The distance between the sensitive volumes of ^1H and ^{19}F is 1300 μm .

(1:2.5). The Gel-BmimDCA sample contains 2.5 ml of BmimDCA.

A single-sided NMR scanner operated at a ^1H Larmor frequency of 18.7 MHz, with a static magnetic field gradient of 21.6 T/m. was employed. More details are included in [Supplementary Contents](#).

Due to the strong gradient and the frequency of excitation, both ^1H and ^{19}F nuclei fulfill the condition $\omega_0 = \gamma_H B_0 = \gamma_F (B_0 + \Delta B_0)$ inside the field of view of the scanner, above the radiofrequency (RF) coil (see Fig. 1), with $\Delta B_0 = B_0 (\gamma_H/\gamma_F - 1)$. In this way, a selection of either ^1H or ^{19}F can be made when the thickness of the film is less than the distance between the location of ^1H and ^{19}F resonances. The height of the effective sensitive volume of the device is around 240 μm . In Fig. 1, the regions corresponding to the ^1H and ^{19}F resonance frequency are shown with the values of B_0 and $B_0 + \Delta B_0$, respectively. These values of the field change in the sample when the magnet is moved down. The corresponding ^1H Larmor frequency is $\omega_0/2\pi = 18.7$ MHz. The film, together with the platform remain fixed. In the left-hand side of the drawing, only the fluorine nuclei inside the sample are at resonance with the frequency ν_0 . On the right side, the magnet is moved downwards, allowing only the proton nuclei in the sample to be at resonance with ν_0 . The vertical arrows indicate the distance between the position of the sensitive volume and the RF coil for ^1H and ^{19}F nuclei, in the range of 3.0 mm and 1.7 mm, respectively.

3. Results and discussion

Fig. 2a shows the profile of the sample gel-EmimTf2N (1:1.5), from which a thickness of around 750 μm can be obtained. The ratio of the area under the profiles scales approximately with the number of ^1H and ^{19}F atoms in the IL molecules (11–6, respectively). In order to control whether the scanner is introducing any error in this measurement, a profile of the bulk IL EmimTf2N was measured, filling all the volume on top of the scanner. This profile shows up as a step-wise function proportional to the number of nuclei in the sample (Fig. 2b). A correlation between the signal-proportions and the number of nuclei is obtained. This result is not taking into account possible deviations due to the fact

Table 1

Relaxation times and diffusion values of the sample gelatin-EmimTf2N (1:1.5). The values of the distances are relative to the bottom of the sample holder (see Fig. 2a). For example, the heights 2000 μm and 1400 μm represent the same height in the sample, encoded in two different sensitive volumes corresponding to ^{19}F and ^1H nuclei. The last two rows show the bulk values of EmimTf2N.

Distance from bottom surface (μm)	T_2 (ms)	T_1 (ms)	$D(10^{-11} \text{ m}^2/\text{s})$
2600 (^{19}F)	347 ± 31	694 ± 62	3.7 ± 0.2
2300 (^{19}F)	389 ± 35	640 ± 32	4.2 ± 0.2
2000 (^{19}F)	412 ± 37	706 ± 66	3.4 ± 0.2
1400 (^1H)	388 ± 27	483 ± 39	4.9 ± 0.2
1100 (^1H)	325 ± 16	470 ± 19	5.4 ± 0.1
800 (^1H)	306 ± 15	517 ± 21	5.0 ± 0.1
Bulk EmimTf2N (^1H)	409 ± 10	431 ± 8	5.0 ± 0.1
Bulk EmimTf2N (^{19}F)	656 ± 10	763 ± 10	4.1 ± 0.1

that (i) the signal is related to the specific NMR sensitivity of the atom and, (ii) in NMR unilateral sensors there is a dependence of the B_1 power to the distance of the sensitive volume from the coil.

In order to study the dynamics and effect of interaction between the molecules of IL and gelatin in the sample, relaxation and diffusion measurements were performed at three different heights in the film. The results are summarized in Table 1. Due to the relatively large error in the relaxation time values, no significant differences of T_1 and T_2 can be distinguish in different layers of the film.

The diffusion coefficients were measured at a diffusion time of 2 ms. No systematic dependence on depth is found. No significant differences between the confined and the bulk values at this diffusion time are found.

Fig. 3 shows the profile of the same material with a different ratio, sample gel-EmimTf2N (1:2.5), with a thickness of around 340 μm , and the diffusion coefficients as a function of the diffusion time. In the center of the film, the relaxation times are $T_2 = (327 \pm 9)$ ms for the

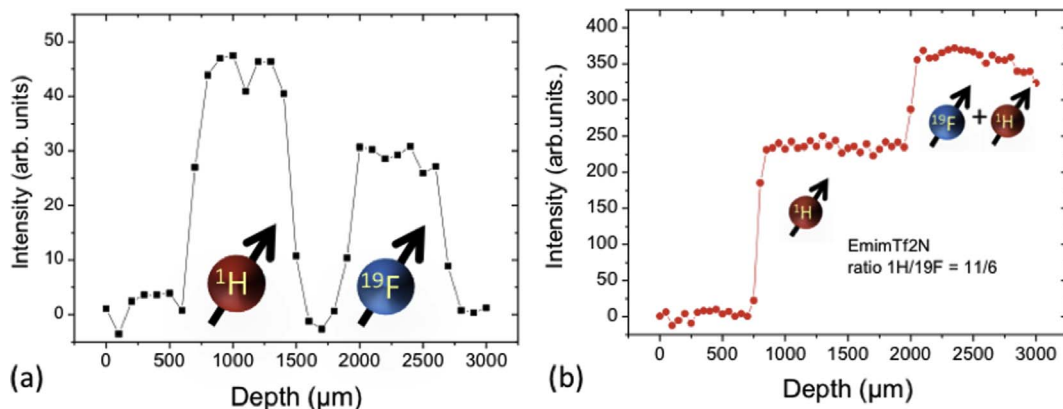


Fig. 2. (a) Profile of the sample Gel-EmimTf2N (proportion gel-IL 1:1.5). (b) Profile test measurement of the bulk IL EmimTf2N. Below 2000 μm , only the ^1H signal is detected.

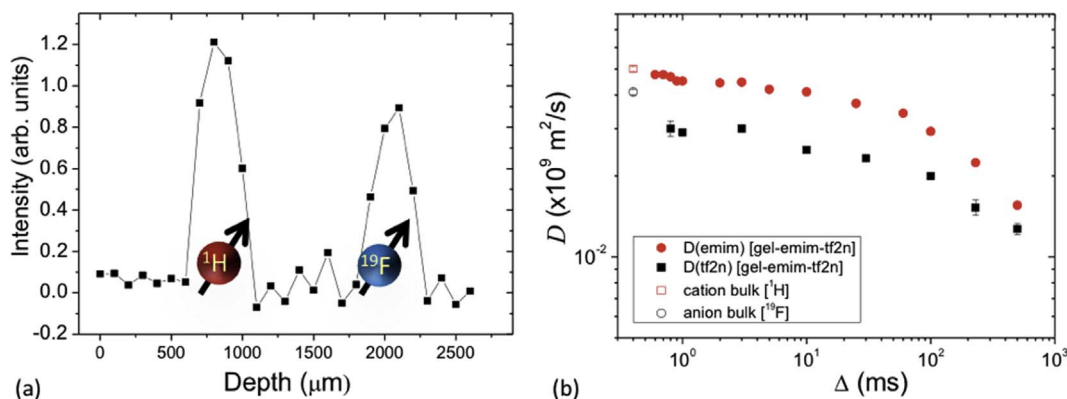


Fig. 3. (a) Profile of the sample Gel-EmimTf2N (proportion gel-IL 1:2.5). (b) Diffusion values in function of the diffusion time Δ . The measurements are performed in a layer of 100 μm in the center of the film.

Table 2

Relaxation times and diffusion values of the bulk EmimBF₄.

	T_2 (ms)	T_1 (ms)	$D(10^{-11} \text{ m}^2/\text{s})$
Cation (^1H)	245 ± 4	475 ± 4	5.3 ± 0.1
Anion (^{19}F)	394 ± 6	790 ± 10	4.8 ± 0.1

anions. Two components are found for the cations, $T_{2,\text{short}} = (0.50 \pm 0.07)$ ms and $T_{2,\text{long}} = (278 \pm 5)$ ms, with relative weights of 18% and 82%, respectively, both values being somewhat shorter than those for the previous sample. The spin-lattice relaxation times in this film are $T_1 = (580 \pm 20)$ ms for the anions and $T_1 = (419 \pm 13)$ ms for the cations.

All samples of gelatin-EmimTf2N are a compact rigid solid white material. However, following from the relaxation and diffusion measurements, the IL retains most of its bulk dynamical properties inside the pore network constituting the gelatin-film at short times. No significant geometrical restriction in the dynamics is detected at diffusion times below 2 ms. On the other hand, the electrical conductivity values of the bulk EmimTf2N and the gelatin-EmimTf2N differ from each other by three orders of magnitude, being $\sigma = (0.61 \pm 0.06)$ S/m for the bulk IL and $\sigma = (5.0 \pm 0.6)10^{-4}$ S/m for both films. This shows a strong deviation between a macroscopic and microscopic transport properties represented by conductivity and diffusion, respectively. An indication of this behavior is seen in the dependence of the self-diffusion coefficients as a function of time, which decreases about threefold when increasing the diffusion time from 1 ms to 500 ms, the longest value experimentally accessible due to the limitations by T_1 . For the shortest and longest diffusion times, a root-mean-squared displacement on the order of 0.2–4 μm can be computed, suggesting that significant

Table 3

Relaxation times values for the cation in the sample Gelatin EmimBF₄. The values in brackets indicate the relative weights of the components. All relaxation values are measured at the center of the film.

	$T_{2,\text{short}}$ (ms)	$T_{2,\text{long}}$ (ms)	T_1 (ms)
Cation (^1H)	0.9 ± 0.2 (20%)	246 ± 7 (80%)	448 ± 8
Anion (^{19}F)	0.7 ± 0.1 (18%)	321 ± 9 (82%)	650 ± 20

restrictions to the translational ionic mobility occurs on length scales of 1 μm or below.

The IL in the sample Gel-EmimBF₄, consists of the same cation but a different anion, BF₄. The NMR parameters of the bulk EmimBF₄ IL at room temperature are presented in Table 2. The diffusion values are in agreement with those found in the literature [11].

The profile of the film (Fig. 4a) shows a thickness of around 450 μm . Under confinement, the CPMG decays for both ions could be fitted with two exponential decays. The results are summarized in Table 3. The diffusion values were determined in a wide range of diffusion times (see Fig. 4b). As in the case of gel-EmimTf2N, the diffusion constants have similar values that in bulk for short times. However, a much more pronounced time-dependence is found, leading to a reduction of the self-diffusion coefficient by about a factor of 15 at 500 ms. On the other hand, the ionic conductivity is reduced by more than three orders of magnitude: only an upper limit can be given in the film ($\sigma < 0.0005$ S/m), while the bulk value in EmimBF₄ is $\sigma = 1.25$ S/m [13].

Using BmimDCA, a sample with a very different texture was obtained. The visual and macroscopic properties of this Gel-BmimDCA sample are characteristic of the so-called ion-jelly type, being a flexible and semitransparent gel. This is related with the high hydrophilicity of

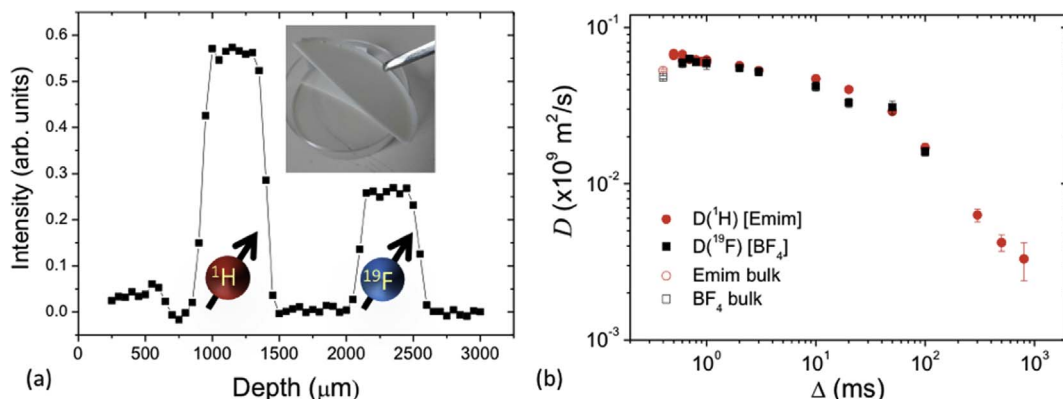


Fig. 4. (a) Single sided NMR profile of the sample Gel-EmimBF₄. The inset shows a photograph of the sample. (b) Diffusion values in function of the diffusion time Δ . The measurements are performed in a layer of 100 μm in the center of the film.

Table 4

Averaged relaxation times and diffusion values for the cation of BmimDCA in the film and bulk samples. The diffusion time is 2 ms. The values in brackets indicate the relative weights of the components.

	$T_{2,short}$ (ms)	$T_{2,long}$ (ms)	T_1 (ms)	$D(10^{-11} \text{ m}^2/\text{s})$
Film	1.3 ± 0.2 (20%)	32.1 ± 0.4 (80%)	94 ± 2	1.5 ± 0.2
Bulk	–	188 ± 1	234 ± 7	5.2 ± 0.1

this IL [16]. The profile from the ^1H signal (figure A1a, Supplementary Content) shows a relatively homogeneous ^1H spin density along the sample, with only the cation generating an NMR signal. The results of the echo decays in the diffusion measurements are shown in Figure A1b. The CPMG echo decays could be fitted with two components. The results are summarized in Table 4. All the NMR parameters demonstrate that the dynamics of this IL is very restricted inside the film. A remarkable fact in this case is that the electrical conductivity measured in the bulk IL, and inside under confinement in the film, show the same value, $\sigma = (0.26 \pm 0.03) \text{ S/m}$. This indicates a decoupling of ionic self-diffusion and electrical conductivity in the sample.

In the films with ILs containing Emim as cation, the gelatin forms a “hard”, rigid matrix confining the ILs. Since NMR is sensitive to all of the protons in the structure of the samples, the two components of T_2 observed in both the cation and anion in gel-EmimBF₄, indicates that one fraction of the IL molecules have very low mobility, taking part in the gel-matrix structure and possibly being at least intermittently bound to the partially renatured collagen molecules of the gelatin [14]. The residual water is tightly bound to the gelatin by hydrogen bonds, and the NMR signal is expected to decay significantly before the first echo is formed. In the same way, in the gel-EmimTf₂N, a short component is not observed, because it decays before the first echo is formed. The values of the longer T_2 component and T_1 , comparable to the bulk values, give evidence of fast rotational and translational [12] dynamics. However, the ionic conductivity does not correlate directly with the microscopic dynamics. The time dependence of the diffusion coefficient, however, shows restriction in the translational dynamics, at times longer than few milliseconds. Below ~ 2 ms, translational molecular motions become apparently faster than in bulk for the case of EmimBF₄. An increase of ionic mobility of ILs in confinement is indeed reported in literature; for example, EmimBF₄ confined in surface modified (silanized) silica nanopores, has a higher diffusion coefficient than in bulk, because of changes in the ion packing due to a reduction of the density of the IL in small pores [15]. The variation of diffusion with time gives evidence that in these gel-IL samples, the gelatin matrix has characteristics of a porous material, where the solvent is moving within the geometrical restrictions of the pore walls.

In the case of gel-BmimDCA, the hydrophobic chain of the cation interferes with the hydrogen bonding network of the gelatin triple helix and the presence of the large ions bound to the gelatin strands prevent the formation of the triple superhelix, giving more flexibility to the film [16] and hindering the translational mobility of the IL molecules. However, the bulk-like ionic conductivity is retained. As explained in the literature [17], in this type of non-rigid gel-matrix, new ion-conducting pathways are possibly created and playing an important role in ionic conduction at the microscopic and/or quasi-macroscopic scale.

4. Conclusions

The central subject of this study was to determine the ionic mobility of ILs inside gelatin films. The results show that the nature of the ILs is critical in the final mechanical characteristics of the gelatin films, having influence in the mobility of the confined ions. When the gel-film is a rigid brittle material, the rotational dynamics were not affected due to confinement effects and geometrical restrictions of the gel matrix of gelatin, on the time scale of few milliseconds. However, translational mobility is affected at longer time scales, up to 500–800 ms. Electrical conductivity in this case is severely reduced. In the case of gel-BmimDCA films, the macroscopic electrical conductivity was observed to be similar as the conductivity of the bulk IL. This study opens new possibilities toward the design of novel and stable gel-conductors for different electrochemical devices. Considering ILs as electrolytes, this investigation could have impact in the field of energy storage devices like batteries or electroactive polymeric materials.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.micromeso.2017.11.050>.

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